OPEN AGENT HACKATHON 2026 · REASONING ARCHITECTURE

Pulsar 5

A disaster response planning agent that **changes its mind correctly** when the world changes.

> Hurricane Ian, September 2022. Early forecasts pointed at Tampa Bay. Landfall came 100 miles south, at Fort Myers. This is a system built for that gap.

THE PROBLEM

Seven systems, none of them agreeing, all of them moving

Weather

A new advisory every six hours. The cone moves.

Shelters

Capacity reported hours late. One fills early and the plan is wrong.

Population

Who cannot drive, who is over 65, who is on oxygen.

Roads

A ramp closes and a staging site is suddenly unreachable.

Decisions have to be made fast, revised as the storm moves, and **defended afterwards** — to an inquiry, to a county commission, to the press.

THE USER

The county EOC duty officer

What they decide, in the next 24 hours

- Which shelters to open, and when
- Where to stage supplies, with a fixed truck fleet
- Which zones to evacuate, and in what order

What makes it hard

- An evacuation order cannot be taken back
- The forecast will change, probably twice
- Every call gets reviewed afterwards

Pulsar 5 does not replace this person. It hands them a ranked plan where **every line cites its evidence**, and it refuses to commit anything irreversible on their behalf.

THE TEST CASE · HURRICANE IAN, SEPTEMBER 2022

Advisory 15 · 26 Sep

Tampa Bay

The forecast cone covered Pinellas and Hillsborough. Counties north prepared.



Landfall · 28 Sep

Fort Myers

Lee County, about 100 miles south. A Category 4 at the coast.

> Every plan built on advisory 15 was **reasonable, evidence-backed, and aimed at the wrong county**.

What the gap cost

NHC Tropical Cyclone Report AL092022

$112.9B
total US damage — third-costliest hurricane on record

156
deaths in the US, 66 of them direct

36 of 41
storm-surge deaths were in **Lee County** alone

52,514
structures impacted in Lee County, 5,369 destroyed

The dead skewed old

The report notes the age distribution peaks toward the elderly, "consistent with other hurricane landfalls where the oldest die at the highest rates." Of the 90 indirect deaths, the leading cause was **lack of access to timely medical care**.

This is why Pulsar 5 ranks zones on over-65 share, no-vehicle households, disability and poverty — not on population alone.

And it keeps happening. **Helene** (2024) — $79.6B, 219 deaths. **Milton** (2024) — $34.3B, 32 deaths. Over $100B across six states in two weeks.

The other half of the problem is over-warning: roughly **417 miles** of coastline warned per storm, with hurricane-force winds reaching about a quarter of it.

Sources: NHC Tropical Cyclone Report AL092022 · NOAA NCEI billion-dollar disasters · Frontiers in Public Health 7:149 (2019)

WHY THIS IS AN AGENT PROBLEM, NOT A FORECASTING PROBLEM

The forecast was not wrong. The plans built on it were stale.

> “the cone of uncertainty did contain the landfall location for **all advisory cycles**”
>
> — National Hurricane Center, post-storm report on Ian

The forecast was normal

Ian's 48-hour track error was **67.8 n mi** — about 78 miles, in line with the preceding five-year average. At 24 hours it beat the average.

The surge forecast tripled

At Fort Myers Beach it went **4–7 ft → 8–12 ft → 12–18 ft** in about a day as the track slid south. Measured peak: 12.70 ft.

30 miles decided everything

Pirate Harbor measured **4.47 ft**. Sanibel, under 30 miles away, **12.58 ft**. Tampa Bay sat **4.38 ft below** normal.

Every advisory carried the information needed to protect Lee County. Nobody was missing data — the plans simply **outlived the facts they were built on**. You cannot forecast your way out of that. You have to replan.

Source: NHC Tropical Cyclone Report AL092022, Forecast and Warning Critique · Tables 5a, 7

THE SOLUTION

The model proposes. Code and a critic decide.

01

Grounded

Every fact enters through Zetaris and is stored with its source table, its SQL and its timestamp.

02

Simulated

Capacity, coverage and distance are checked in plain code. No model gets a vote on the verdict.

03

Criticised

A Critic checks each claim against the evidence it cites, and can send the plan back.

04

Gated

Anything irreversible goes to a human. Always, regardless of how confident it is.

And when a new advisory lands, it **invalidates only the decisions that cited a fact that actually changed** — then replans those.

Architecture

six agents · one shared state · three engines

ZETARIS — the only data path

storm_advisories · shelters · hospitals · acs_vulnerability · staging_sites · road_events · disaster_declarations

WORLD MODEL — shared state every agent reads and writes

beliefs with provenance · decisions with dependencies · conflict log

Analyst
tool loop over Zetaris

Planner
authors every action

Simulator
plain code, no model

Critic
can reject the plan

Risk
the human gate

Belief Reviser
targeted replan

Planner ⇄ Simulator ⇄ Critic iterates, bounded at 3. H-MEM carries lessons across runs; Markovian keeps the hourly replay flat.

Six agents, six different jobs

Agent	Lane	What it actually does	Hands off to
Analyst	fast	Picks queries from a fixed menu, runs them through Zetaris, writes each row in with its provenance	Planner
Planner	deep	Authors the plan. The only agent allowed to create actions, and every one cites fact ids	Simulator
Simulator	code	Capacity, coverage, distance, budget. Returns structured violations, never prose	Critic
Critic	judge	Four mechanical checks plus a model review. Can reject and send instructions back	Planner, to revise
Risk	fast	Reversible or not. An evacuation order never auto-commits, at any confidence	The human queue
Belief Reviser	deep	Diffs the world, walks the dependency graph, invalidates exactly the dependents	Planner, targeted

Each lane is a separately pinned Nemotron route through Nightshift. The Simulator's verdict is the one thing no model can override.

It is a loop, not a chain

from a real trace, run ian-landfall-minus-48h

ITERATION 1 — REJECTED

Planner opens 3 shelters — opening one costs staff and cots, so it starts lean.

Simulator 14,184 expected evacuees against 5,441 open spaces. Short by 8,743.

Critic refuses at confidence 0.56 and sends the shortfall back as an instruction.



ITERATION 2 — ACCEPTED

Planner raises the opening budget for exactly the reason given, and reaches 23,556 spaces.

Critic accepts at 0.78. **Risk** then holds 11 of 29 actions for a human and lets 18 commit.

The first pass is a genuine planning trade-off, not a staged failure. Opening every shelter immediately is not free, so the system starts conservative and lets the simulator prove it wrong.

The change of mind

POST /events — advisory 23 arrives

BEFORE · advisory 15, Tampa cone

Hillsborough 3 · Pinellas 3 · Sarasota 3 · Charlotte 2

Lee County: no action

AFTER · advisory 23, Fort Myers cone

Lee 3 · Charlotte 3 · Sarasota 2 · Pinellas 1

Hillsborough: stood down

27
beliefs changed

11
decisions invalidated

36
decisions preserved

+4 -6
actions added, removed

Sarasota was in both cones, so nothing touched it. Thirty-six decisions survived because no fact they cited actually changed. That is the difference between revising and starting over.

Three Meterless engines, each load-bearing

World Model

The shared state. Beliefs carry provenance, decisions carry dependencies. A late write with an older timestamp is logged as a conflict, never allowed to clobber fresher data.

```
8 threads × 60 writes
zero lost
```

H-MEM

Memory across runs, in three levels. A recalled lesson writes a capacity derate onto a shelter, which both Planner and Simulator read — so it changes the plan, not the prompt.

```
run 2 plans differently
because of run 1
```

Markovian

The hourly replay. Each hour is one bounded step with a compressed carryover; H-MEM is injected once, at step one, per the COMBINE contract.

```
24 steps, context flat
at 448–452 chars
```

The Markovian number is the one to check: each step hands back **20 KB** of notes and issues, and the context the next step sees stays at about **450 characters**, from hour one to hour twenty-four.

SPONSOR TECHNOLOGY

Zetaris is the only way data enters

One access point

Seven sources, all registered as Lightning REST tables and queried through POST /sql-editor/sqls/run. Nothing in the app opens a file or calls FEMA, NWS or Census directly.

Registration is scripted

Eight SQL files and a registrar that dry-runs by default and sends **one statement at a time**. No manual setup in the Zetaris UI.

Provenance travels

Each row becomes a belief carrying its source table, the exact SQL, and an as_of timestamp — which is what lets the Critic flag staleness per source.

One consequence worth naming: the NWS alerts API needs a `User-Agent` header that a Lightning REST table cannot send. Rather than bypass Zetaris, we moved to the Iowa Mesonet archive, which needs no header and reaches back to 2002 — the right source for a 2022 replay anyway.

SPONSOR TECHNOLOGY

Nightshift is the only model provider

Check it by grepping

There is no `openai`, `anthropic` or NVIDIA client anywhere in the application. Every call is an OpenAI-compatible POST to the Nightshift Runtime URL, with the `meterless-app` header set.

The NVIDIA key lives only inside the Nightshift app. This process never reads it.

Three Nemotron lanes

fast — Analyst tool calls, Risk classification

deep — Planner, Belief Reviser

judge — Critic

Each pinned separately, so a cheap lane never silently does the Critic's job.

every request records the model that served it — logs/trace—<run_id>.jsonl


```
{"agent_name":"Critic","action":"reject_plan",
 "model":"nvidia/nemotron-3.5-bolt-e4b","lane":"judge",
 "confidence":0.56,"status":"rejected"}
```

Reliability scoreboard

`35 perturbed scenarios · python scripts/eval.py`

0	0	3/3	129	1.76
constraint violations left unhandled	irreversible actions auto-committed	correct abstentions on missing data	plans the Critic caught before commit	mean iterations to an accepted plan

Abstention is a measured behaviour

Lose a **core** source — the advisory, the zones, the shelters — and the run declines to plan rather than inventing a number. Lose an **optional** one and it degrades: the plan is made, and the trace records what was missing.

34 tests, and a fresh-clone check

Engine conformance, loop behaviour, escalation rules. Cloned fresh with an empty `.env` and both services stopped, a full 29-action plan still comes back in 5 ms.

What this is not

stated in the README, not buried

Not an official decision tool. A hackathon prototype. Nobody should evacuate a county on its say-so.

Shelter occupancy is simulated. No real shelter census or patient data appears anywhere in the repository. Facilities are synthetic, to FEMA-shaped distributions.

The 25-mile buffer under-covers on purpose. A Category 4 wind field is far wider. The system surfaces those zones to the operator instead of ordering evacuations there.

The vulnerability score is deliberately crude. Four ACS signals, equal weight, chosen to be auditable rather than optimal. It is not CDC SVI.

Routing is straight-line distance, not road network distance. Road closures in the replay are synthetic and labelled as such.

Scoped to five Florida counties and one storm. The code takes any county list; the data behind Zetaris does not. Going wider means FIPS codes, not county names.

An emergency manager deserves to know exactly where the model stops being trustworthy. Writing that down is part of the product, not an apology for it.

PULSAR 5

At advisory 15, with every forecast pointing at Tampa, it had already flagged Lee County to the operator.

```
distant_vulnerable_zone: Zone FL-071-0500 (Lee) scores 0.83 on social
vulnerability but lies 48 mi outside the current cone. No evacuation is
warranted yet; pre-positioning there is an operator call.
```

Not a prediction. The Critic simply refused to let a high-vulnerability area disappear just because the current cone missed it — and handed the judgement to the human, where it belongs.

github.com/lazizbekravshanov/pulsar5